Victor M. Renteria

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EDUCATION

University of California Irvine - Irvine, CA Graduation: Mar. 2027

B.S. Software Engineering

- Relevant Coursework: Python Programming, Data Structure Implementation and Analysis, Human Computer Interaction, Requirements Analysis and Engineering, Design and Analysis of Algorithms, Computer Networks, Compilers and Interpreters, Embedded Software
- Technical Skills: Python, C++, Java, SQL, React Native, Node.js, Firebase, Tensorflow, Prisma, PostgreSQL
- Awards: Chancellor's Excellence Scholarship ($12,000), CLLCF Scholarship ($6,500), Dean's List

Projects

Early Autism Screening Mobile App - Irvine, CA Sep. 2025 - June 2026

Software Engineer - Senior Capstone Project

- Developed a cross-platform mobile application for early autism screening using React Native and Expo SDK 54, backed by an Express.js REST API and PostgreSQL database managed via Prisma ORM.
- Designed and implemented a biometric data pipeline that samples heart rate at 1Hz during multi-video screening sessions, persists timestamped readings to AsyncStorage, and renders post-screening analytics with avg/max/min BPM statistics
- Configured Firebase Authentication with Google Sign-In for both iOS and Android, handling platform-specific provisioning, SHA-1 certificate registration, and OAuth redirect schemes

FluentHands - Irvine, CA Feb. 2026

IrvineHacks participant

- Developed a "Duolingo for ASL" web app using React and Tailwind CSS, allowing users to learn the 26-letter ASL alphabet through interactive gamified modes.
- Integrated MediaPipe and TensorFlow to detect 21 hand landmarks, training a neural network to classify gestures with a fallback to custom logical finger-positioning algorithms.
- Achieved 92% gesture recognition accuracy across 26 cases from 10 users.

AI Powered Fitness Platform - Irvine, CA Nov. 2025

ZotHacks participant

- Architected a full-stack web application that generates personalized workout plans by integrating the OpenAI API, processing user biometrics (height, weight, age) and fitness goals.
- Developed a robust RESTful API using Node.js and Express to handle asynchronous AI requests, ensuring immediate user feedback by decoupling database persistence from LLM response times
- Implemented a relational database schema with Prisma ORM, managing complex data relationships between user profiles, workout plans, and historical activity logs.

Experience

Robotics Club - Whittier, CA Sep. 2024 – Dec. 2024

Embedded Systems Developer

- Assembled and launched an unmanned rocket with a custom electronic payload, developing C++ software to capture environmental data including pressure, temperature, humidity, and 9-axis motion.
- Skillfully soldered and integrated electronic components, ensuring reliable data transmission and performance under launch conditions.
- Designed and 3D-printed a durable payload container with Autodesk Fusion, protecting sensitive electronics during high-impact launch and recovery phases.

Cal Poly Pomona - Pomona, CA

University Researcher June 2023 – Aug. 2023

- Co-authored and presented research evaluating synthetic vs. real-world training data for object detection, contributing to literature on multi-view 3D reconstruction techniques including Structure-from-Motion (SfM)
- Engineered an end-to-end synthetic data pipeline using RealityCapture, Blender, and Unity, converting SOLO datasets to COCO format to train a custom YOLOv object detection model, achieving **66.1% mAP, 77.7% precision, and 62% recall** on the GMU Kitchen Dataset
- Developed 3D photogrammetry workflows using a camera, turntable, and photo studio to reconstruct textured mesh models, supplemented by BigBird Dataset assets, reducing reliance on costly real-world data collection

MESA & TRIO Student Support Services STEM - Whittier, CA Aug. 2023 – Aug. 2024

Tutor

- Tutored math and physics mainly, Calculus II: Provided individualized support to over 10 students with varying abilities, enhancing their understanding of advanced calculus concepts, and improving overall class performance.
- Developed and Delivered Workshop Curriculum: Designed a structured Calculus II workshop in collaboration with professors, attended by 12 students, achieving a 100% pass rate among participants.

SKILLS AND INTERESTS

Languages: English, Spanish
Interests: Soccer (Real Madrid), Jiu Jitsu, Muay Thai, Chess, Films (Drama), Cooking (Steak), Music (Alternative Rock, Indie, Rap)
Affiliations: MESA, SHPE, Lambda Upsilon Lambda Fraternity Incorporated, National Honor Society